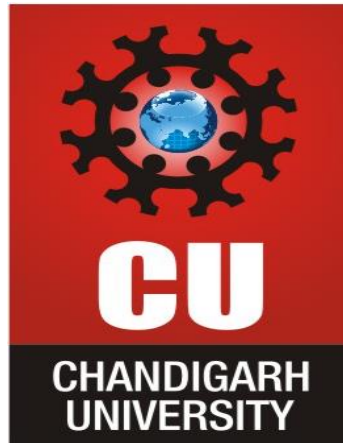




**EXPERIMENT : 06**

**NAME :** Manan Verma

**SEMESTER :**5th

**SECTION :**24\_AIT\_KRG-G1-B

**UID :**24BAI70008

**SUBJECT CODE :**24CSP-310

**SUBJECT NAME :**Advanced Database Management  
System

**FACULTY'S NAME :** Mr. Alok Kumar

[A.]

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

Sales

| <u>sale_date</u> | fruit   | sold_num |
|------------------|---------|----------|
| 2020-05-01       | apples  | 10       |
| 2020-05-01       | oranges | 8        |
| 2020-05-02       | apples  | 15       |
| 2020-05-02       | oranges | 15       |
| 2020-05-03       | apples  | 20       |
| 2020-05-03       | oranges | 0        |
| 2020-05-04       | apples  | 15       |
| 2020-05-04       | oranges | 16       |

Output

| <u>sale_date</u> | diff |
|------------------|------|
| 2020-05-01       | 2    |
| 2020-05-02       | 0    |
| 2020-05-03       | 20   |
| 2020-05-04       | -1   |

Ans.

**SELECT**

**sale\_date,**

**SUM(CASE WHEN fruit = 'apples' THEN sold\_num ELSE 0 END)**

**-**

**SUM(CASE WHEN fruit = 'oranges' THEN sold\_num ELSE 0 END) AS diff**

**FROM Sales**

**GROUP BY sale\_date**

**ORDER BY sale\_date;**

[B.]

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- **id** represents the current node.
- **p\_id** represents the parent node.

Classify every node into one of the following categories:

- **Root:** A node that does not have any parent.
- **Leaf:** A node that has a parent but does not have any children.
- **Inner:** A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

| Tree |      | Expected Output |       |
|------|------|-----------------|-------|
| id   | p_id | id              | Type  |
| 1    | NULL | 1               | Root  |
| 2    | 1    | 2               | Inner |
| 3    | 1    | 3               | Leaf  |
| 4    | 2    | 4               | Leaf  |
| 5    | 2    | 5               | Leaf  |

Ans:

**SELECT**

**id,**

**CASE**

**WHEN p\_id IS NULL THEN 'Root'**

**WHEN id IN (SELECT p\_id FROM Tree) THEN 'Inner'**

**ELSE 'Leaf'**

**END AS Type**

**FROM Tree**

**ORDER BY id;**